INCH-POUND

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MILITARY SPECIFICATION

TELESCOPE, ELBOW: 9356181

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for the 4-power fixed focus elbow telescope which is the optical portion of Sightunit: M67. Throughout this specification this instrument is referenced as "telescope." This telescope has a radioactively illuminated reticle (see 3.2.1 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-O-13830 - Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

MIL-F-13926 - Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BACS, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8565556 - Pressure Tester Two Station
9356143 - Reticle, Optical Instrument
9356181 - Telescope, Elbow: 9356181
IEL 9356181 - List of Inspection Equipment Documents for Telescope, Elbow: 9356181

PACKAGING DATA SHEET

PS 9356181 - Packaging of Telescope, Elbow: 9356181

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DLAM 4145.8 - Radioactive Commodities in the DOD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for Protection Against Radiation
Part 30: Rules of General Applicability to Licensing of By-product Material

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9356181, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the manufactured telescope shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking
d. Workmanship.

3.2.1 Radioactive material caution. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling, and shipping of radioactive material.

3.2.2 Eyeshield. The rubber eyeshield shall show no evidence of deterioration, cuts or tears and shall fit snugly on the telescope body.

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3.3 Environmental service conditions. Any telescope subjected to any or all of the requirements of 3.3.1, 3.3.2, 3.3.3, and 3.3.4 shall subsequently meet the requirements of 3.3.5 through 3.4.7 unless otherwise specified.

3.3.1 Vibration. The telescope shall be positioned with the line of sight through the eyepiece and objective tube in a horizontal plane and the telescope supported at the .8435 inch diameter of the objective tube. The telescope shall then be vibrated at a constant frequency of 30 ± 5 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

3.3.2 Shock. The telescope shall be shocked six (6) times in each of two (2) positions. The applied shocks shall be in a vertical, up direction. In the first position, the telescope shall be positioned with the line of sight through the eyepiece and objective tube in a horizontal plane. In the second position, the telescope shall be positioned with the line of sight through the eyepiece and objective tube in a vertical plane. In both positions the telescope shall be supported at the .8435 inch diameter of the objective tube. The shock, for all twelve (12), shall be 300 g's for one (1) millisecond and have a 1/2 sine wave pulse.

3.3.3 Temperature. The telescope shall not suffer permanent damage when subjected to the constant temperature extremes of -50°F (-46°C) for a period of 24 hours and +160°F (+71°C) for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 when returned to room ambient.

3.3.4 Sealing and flushing. The telescope shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5 ± 0.10 psig for a period of two hours using dry nitrogen having a dew point at least as low as -25°F (-32°C). The telescope shall then be flushed with dry nitrogen until the interior atmosphere of the telescope has a dew point at least as low as -25°F (-32°C).

3.3.5 Condensation (low temperature). When viewed through the eyepiece and objective ends of the telescope, there shall be no evidence of condensation at room temperature immediately upon thermal stabilization, subsequent to being subjected to the -50°F (-46°C) and prior to the +160°F (+71°C) requirement of 3.3.3.

3.3.6 Cleanliness. The telescope shall meet the cleanliness requirements of MIL-O-13830. When viewed through the eyepiece, dirt shall not exceed the following limitations of dirt, quantity, and size (see 6.6).

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a. Inner zone (100 mils diameter). The maximum size of any one particle of dirt or foreign material shall not exceed the area equivalent to a size 5 dig. There shall be no more than 6 particles of dirt. Three particles may be as large as a size 5 dig. Three particles shall not exceed a size 1 dig. Any particles smaller than a size 1/4 dig shall not be counted.

b. Outer zone (limited to clear aperture of reticle). The maximum size of any one particle of dirt or foreign material shall not exceed the area equivalent to a size 10 dig. Three particles may be as large as a size 10 dig. Smaller sizes may be more in number, but their combined area shall not exceed that of 3 size 10 digs, which is maximum dirt permitted. Any particles smaller than a size 1/4 dig shall not be counted.

c. Dirt separation. The separation between each dirt particle or foreign matter shall be at least 10 mils for both zones.

d. Glass surfaces. All glass surfaces shall be free of condensation, or stains resulting from condensation, when viewed through the objective and eyepiece ends of the telescope.

3.3.7 Defects. There shall be no evidence of moisture, grease, fingerprints, condensations, fractures, adhesive separations, or other forms of optical degradation on any glass component when the telescope is viewed through both the eyepiece and objective ends.

3.3.8 External tritium contamination. Tritium in any form shall not be present on the external surfaces of the telescope in excess of 1000 dpm/100 sq. cm.

3.3.9 Tritium diffusion. Radioactive content of the test bath water shall not exceed 50 nanocuries/day.

3.3.10 Illumination check. Illuminated areas shall be given a visual check to determine if the light source is illuminated.

3.4 Performance.

3.4.1 Collimation. The optical axis of the telescope shall coincide with the geometric axis within .50 mil when the telescope is rotated 3200 mils (90° to the left and 90° to the right from a vertical position of the eyepiece) about the geometric axis as defined by the exterior mounting surface of the objective tube of the telescope.

3.4.2 Parallax. Parallax at the optical axis shall not exceed 0.1 mil when viewing a target between 44 yards (40 meters) and infinity.

3.4.3 Resolution. The resolution on the optical axis shall be 15 seconds of arc or less at one focus setting of an auxiliary observation telescope positioned at the eyepiece.

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3.4.4 Parallelism of reticle and image. When three of the four indices (90 degrees apart) on the telescope body are successively in coincidence with the index inscribed on the body of the applicable telescope mount (see 6.1), the vertical reticle line shall be parallel with the image of a plumb line within 30 minutes of arc.

3.4.5 Reticle accuracy. The angular value for graduations of the reticle shall be as indicated on drawing 9356143 within 3 percent.

3.4.6 Eyepiece focus. The eyepiece focus of the reticle shall be set between -.75 and -1.00 diopter.

3.4.7 Illumination. The reticle lines shall be clearly distinguishable when observed, with and without a 50% neutral density filter, at light levels varied from zero to ten foot-candles.

3.5 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which affect their protective qualities or appearance. Optical surfaces shall be clean and free of any scratches or blemishes which may affect their optical qualities and shall meet the requirements of 3.3.6 and 3.3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg. (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample, as designated by the contracting officer, consists of three telescopes plus three (3) each of all items covered by QAP for inspection in accordance with 4.3.2. The following items are covered by QAPs:

Name	QAP
Eyepiece assembly	SQ 9356136
Lamp, Radioluminous	SQ 9356141
Lamp Assembly	SQ 9356139

4.3.2 Inspections to be performed. First article assemblies, components, and test specimens shall be subjected to all of the examinations and tests specified in Table I (functional tests), Table II (special sampling), and QAPs listed in 4.3.1. It may be inspected for the compliance with any or all other requirements of the applicable drawings or QAPs.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package List (TDPL). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

a. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with DLAM 4145.8 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum

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inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection of item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests.
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Performance	3.4	4.7.2
102. Cleanliness	3.3.6	4.7.1.6
103. Defect	3.3.7	4.7.1.7
104. External Tritium	3.3.8	4.7.1.8
Minor:	None	N/A

4.5 Special sampling. Three telescopes shall be selected at random from each 100 produced, or each month's production, whichever occurs first. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
105. Vibration	3.3.1	4.7.1.1
106. Shock	3.3.2	4.7.1.2
107. Temperature	3.3.3	4.7.1.3
108. Sealing	3.3.4	4.7.1.4
109. Tritium Diffusion	3.3.9	4.7.1.9
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until

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necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of the specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Special inspection equipment. Special testing equipment to test the performance of the telescope shall consist of a telescope holding device and a target projector collimator. The test equipment shall conform to the following requirements:

a. Telescope holding device. This device shall be capable of securing and maintaining the exterior mounting surface of the telescope's horizontal objective tube within 20 seconds of arc. The device shall provide sufficient tension to the objective tube to permit it to be maintained in a "zero" rotational position while the telescope's mechanical axis lays in a vertical plane. This holding device shall also permit the telescope to be rotated through the excursion specified in 3.4.1 and 3.4.4 without slippage, or rotation

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under its own weight. This holding device shall also be equipped with three telescope position indices. One index shall indicate the "zero" rotational position of the telescope within plus or minus 3 minutes, the second index shall be positioned $90^{o} \pm 0^{o}3'$ clockwise from the vertical plane, the third index shall be positioned $90^{o} \pm 0^{o}3'$ counterclockwise from the vertical plane.

b. Target projector collimator. The target projector collimator shall have a clear aperture of at least 1.5 inches. The target reticle of the collimator shall contain a 1.0 mil diameter circle at the intersecting point of the horizontal and vertical reticle lines to determine conformance with collimation, as specified in 3.4.1. Tolerance marks shall also be incorporated on both sides of the vertical reticle line to determine conformance to parallelism of reticle and image, as specified in 3.4.4. Unless otherwise specified, the target collimator shall be set at infinity focus. The target collimator shall be positioned and aligned so that its optical axis is adjacent to, and parallel within 40 seconds to the geometric axis of the objective tube of the telescope under test. The vertical reticle line of the target collimator shall be adjusted plumb.

c. Sealing. Special testing equipment to test sealing of the telescope shall conform to the internal pressure test requie-ments of Drawing 8565556, Pressure Tester - Two Station. A special adapter shall be utilized when applying, or flushing, the nitrogen gas in the telescope.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers, and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. Conduct the vibration test with the telescope positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitude and frequency specified in 3.3.1. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle, the telescope shall be removed from the vibration adapter and subjected to an external tritium contamination test as specified in 4.7.1.8 and then to a visual, tactile and audible examination as specified in 4.7.1.6 and 4.7.1.7. The telescope shall then be subjected to the tests in 4.7.2 to determine overall conformance to the requirements of 3.4.

4.7.1.2 Shock. Conduct the shock test with the telescope positioned on an appropriate adapter affixed to a shock test machine that is capable of providing the shock as specified in 3.3.2. The adapter shall be capable of holding and positioning the telescope as specified in 3.3.2. The shock pulse shape, peak value and normal

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time duration shall be as specified in 3.3.2. Subsequent to the 12 shocks the telescope shall be removed from the shock test adapter and subjected to an external tritium contamination test as specified in 4.7.1.8 and then to a visual, tactile and audible examination as specified in 4.7.1.6 and 4.7.1.7. The telescope shall then be subjected to the tests in 4.7.2 to determine overall conformance to the requirements of 3.4.

4.7.1.3 Temperature. The telescope shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours ±15 minutes, from room ambient temperature to -50°F (-46°C). The telescope shall be maintained at this temperature for a minimum of 5 hours. Upon completion of the 5 hour exposure the temperature of the chamber shall be raised, over a period of 4 hours ±15 minutes, to room ambient temperature. The telescope shall then be removed from the test chamber and subjected to external tritium contamination test as specified in 4.7.1.8 and then to the visual, tactile, and audible examination as specified in 4.7.1.5, 4.7.1.6 and 4.7.1.7. The telescope shall again be placed in the test chamber and the temperature of the chamber raised, over a period of 4 hours ± 15 minutes, to +160°F (+71°C). The telescope shall remain at this temperature for a minimum of 5 hours. Upon completion of the 5 hour exposure the temperature of the chamber shall be reduced, over a period of 4 hours ± 15 minutes, to room ambient temperature. The telescope shall then be removed from the test chamber and subjected to external tritium contamination test as specified in 4.7.1.8 and then to the visual, tactile, and audible examination as specified in 4.7.1.5, 4.7.1.6 and 4.7.1.7. The telescope shall then be subjected to the tests as specified in 4.7.2 to determine overall conformance to the requirements of 3.4.

4.7.1.4 Sealing and flushing. Internal pressurizing of the telescope to test sealing shall be performed using dry nitrogen gas having a dew point at least as low as -25°F (-32°C) and special testing equipment conforming to 4.6.2.3.1. The telescope shall be pressurized to the requirements of 3.3.4 in accordance with the procedure for internal pressure testing of Drawing 8565556. Maintain the applied pressure for the period specified. With the pressure at 5 ± 0.1 psig, the telescope shall retain this pressure for 2 hours with no indication of leakage in excess of 0.2 psig as indicated on the pressure gage of the testing equipment. Upon satisfactory completion of this test, the body sealing screw shall be immediately assembled to the telescope as depicted on Drawing 9356181 to minimize intermingling of external atmosphere upon disconnection of the pressure tester adapter. The interior atmosphere of the telescope shall conform to the requirements of 3.3.4.

4.7.1.5 Condensation (low temperature). Subsequent to low temperature exposure and immediately upon stabilization at room ambient temperature, the telescope's optical system shall be examined

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as specified in 3.3.5 for evidence of condensation (moisture). This visual examination shall employ the technique of shadowing. The telescope shall conform to the requirements of 3.3.5.

4.7.1.6 Cleanliness. The optical surfaces of the telescope shall be inspected for cleanliness in accordance with MIL-O-13820. Inspection for dirt particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 foot candles. All dirt size evaluations shall take into account the viewing magnification. Cleanliness of the telescope shall conform to the requirements of 3.3.6.

4.7.1.7 Defects. Conduct this test in sequence to the test in 4.7.1.6. Inspect for condensates, grease, fingerprints, fractures, adhesive separations, and etc. This test shall be conducted with the unaided eye utilizing the shadowing technique while viewing through the objective and eyepiece ends of the telescope. The optical system of the telescope shall conform to the requirements of 3.3.7.

4.7.1.8 External tritium contamination. The contamination test shall be performed by wiping all surfaces of the item under test with filter paper moistened with deionized or distilled water. The amount of tritium contamination on the filter paper shall be determined by using a liquid scintillation counting technique. The paper shall be placed in the liquid scintillation solution within one minute after wiping the scope. The liquid scintillation counting system shall have sufficient sensitivity to detect a lower limit of detection of 10 picocuries within 95 percent confidence limit. This test shall be performed by the contractor. The contractor shall furnish filter paper, solution, and bottles. The damp filter paper shall be placed in a container with the proper percentage of scintillation liquid. Insert the sample into this system, count a minimum of 2000 disintegrations (If 2000 disintegrations is not reached in one hour, discontinue the count) and convert the readout into microcuries (222 DPM x 10^4= 1 microcurie). The measured contamination shall be within the limits specified in 3.3.8. The liquid scintillating system must be calibrated, on six-month intervals, to a standard traceable to an N.I.S.T. standard.

4.7.1.9 Diffusion test. Completed scopes, with all luminous vials installed, shall be submerged in a measured volume of distilled or deionized water equal to approximately 10 times the volume of the scope for 24 hours at 23° + 5°C. Scopes shall then be removed from the water. The water shall be analyzed in accordance with the analytical procedures defined below:

a. This is the test procedure for tritium diffusion, and if the radioactive content of the water exceeds 50 nanocuries/day, it shall constitute failure of the test.

b. Analysis of tritium content in the diffusion test shall be made with a liquid scintillation counter. The system calibration shall be established using quenched standards. Total system plus

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standards errors in the standardization shall not be in excess of 10 percent. Efficiencies of the unknown samples shall be established by the channels-ratio method, the external channels-ratio method, or the "H" number method of quench compensation. Counting time shall be established as such that at the test limits, the error (one standard deviation) shall not be greater than 15 percent. The scintillation solution shall be an acceptable commercial water soluble liquid scintillation cocktail. The counting bottles shall be a low potassium liquid scintillation borosilicate glass bottle or polyethylene liquid scintillation vial. When polyethylene scintillation vials are used, a set of quench standards traceable to the National Institute of Standards and Technology shall be made up in the polyethylene vials to determine efficiency. If the quench standards are in glass bottles, a correction factor shall be determined so that the correct results will be obtained from samples in polyethylene scintillation vials.

4.7.2 Performance.

4.7.2.1 Collimation test. This test shall be performed by means of a surface plate, "V" block and a target projector collimator having a clear aperture of at least 2.0 inches. A collimating telescope of at least 3 power (3X) containing a horizontal and vertical reticle line shall also be utilized for alignment of the target projector collimator target reticle.

a. The target reticle of the projector collimator shall contain a 1.0 mil diameter circle at the intersecting point of the horizontal and vertical reticle lines to determine conformance with collimation, as specified in 3.4.1. Tolerance marks shall also be incorporated on both sides of the vertical reticle line to determine conformance to parallelism of reticle image, as specified in 3.4.4. Unless otherwise specified, the target projector collimator shall be set at infinity focus.

b. Target reticle alignment shall be accomplished by placing the 3 power (3X) collimating telescope in the "V" block seated on the surface plate. Direct the line of sight of the collimating telescope toward a plumb line and plumb the vertical reticle line. Position the target projector collimator and establish coincidence of the horizontal and vertical reticle lines on the target projector collimator reticle with the respective crosslines on the collimating telescope reticle.

c. Remove the collimating telescope from the "V" block and position the telescope under test in the "V" block. Observe where the intersecting point of the telescope reticle centerlines falls with respect to the intersecting point on the target reticle. Conformance with 3.4.1 shall be considered acceptable if the intersecting point of the telescope reticle falls within the 1.0 mil diameter circle on the target reticle. This test shall be repeated when the telescope under test is rotated 90° left and again 90° right of the original test position. The intersecting point of the

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telescope reticle shall be within the 1.0 mil diameter circle on the target reticle at both positions to insure overall compliance with 3.4.1.

4.7.2.2 Parallax. This test shall be accomplished by sighting through the telescope of the target projector collimator at the distance specified in 3.4.2. Parallax will be recognized as any apparent movement of the reticle line in relation to the target image when the observer's head is moved from side to side or up and down. Parallax is checked at the optical axis (center of the reticle pattern), and shall not exceed the requirements of 3.4.2.

4.7.2.3 Resolution test. This test shall be performed in accordance with the resolution test in MIL-O-13830, and the telescope shall meet the resolution value specified in 3.4.3.

4.7.2.4 Parallelism of reticle and image test. The test shall be performed in conjunction with the test in 4.7.2.1c. At each of the three (3) positions obtain coincidence of one extremity of the telescope's vertical reticle line with that of the vertical target line. Observe for coincidence at the opposite extremity of the telescope's vertical reticle line. The vertical reticle line of the telescope shall be parallel with the respective reticle line on the target image within the tolerance specified in 3.4.4, as measured on the target. A peep sight placed over the objective of the telescope under test will assist the observer's eye for parallax.

4.7.2.5 Reticle accuracy test. Reticle accuracy test for the angular value of the graduation marks is accomplished by means of a mil block chart with dimensions suitable for the target distance utilized and based on the dimensional notes on drawing 9356143. The error for this test shall not exceed the amount indicated on 3.4.5.

4.7.2.6 Eyepiece focus test. Eyepiece focus test is made with the aid of a 3-power dioptometer adjusted for the individual observer's eye accommodation. Locate the telescope with the objective facing a well illuminated white background so that the reticle pattern is clearly visible. Place the dioptometer in front of the eyepiece and remove parallax by movement of the dioptometer scale. The reading of the dioptometer scale will indicate the error, and shall meet the requirements of 3.4.6.

4.7.2.7 Illumination test. The reticle illumination test shall be performed under simulated conditions specified in 3.4.6. The luminosity test shall be performed in a dark room having a variably controlled light source. To perform this test, the observer must be dark adapted by being in a darkened room for a minimum of 15 minutes before viewing the reticle markings. Prior to performing the test, place a white matted surface approximately 2 feet from the telescope window. Perform the test by observing the reticle pattern and determine that this pattern is clearly discernible by means of the self-illuminating source in the dark room. Gradually raise the level of illumination to where the pattern becomes indiscernible. Center an aperture stop over the objective entrance aperture of the

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telescope and observe that the reticle pattern is clearly discernible. The aperture of the stop shall approximate one half (1/2) the interior diameter of the retainer for the objective lens. Raise the ambient level of illumination to 10 foot candles, remove the fixed stop and observe the reticle pattern. Repeat the above test with a 50% neutral density filter placed over the viewing end of the telescope. The reticle illumination of the telescope, with and without the 50% neutral density filter, shall meet the requirements of 3.4.6.

4.8 Workmanship. The telescope shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The telescope shall conform to the requirements of 3.6.

5. PACKAGING

5.1 General. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet PS 9356181 and DLAM 4145.8. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Telescope, Elbow, 9356181, is supported by Telescope Mount, 9356166, in Sightunit, M67, which is used in connection with direct and indirect fire of weapons.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Requirements for submission of first article sample.
d. Applicable stock number.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC), may also include drawings prepared by,

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and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Mortar
Sightunit
Tritium

6.6 Dirt size equivalent. For comparison purposes, the maximum width of a reticle line is equal to 1.5 dig size.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

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